

Qingqing Li

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EDUCATION

Master of Medicine Xuzhou Medical University, China **Sep 2020 - Jun 2023**

- GPA: 86.30/100
- Thesis: *Microglial Lysosomal Mechanisms for Depressive-like Behaviors Alleviation*

Bachelor of Medicine North Sichuan Medical College, China **Sep 2015 - Jun 2020**

- GPA: 80.06/100 (range: 38/197)

RESEARCH EXPERIENCE

Research Assistant

Kunming Medical University, Kunming, China

Feb. 2026 - May 2026

Principal Investigator: Assoc. Prof. Jiaoqiong Guan

- **Methods:** Integrative genetic analyses (Mendelian randomization (MR), transcriptome-wide association study (TWAS), proteome-wide association study (PWAS), Bayesian colocalization), single-cell transcriptomic analyses (hdWGCNA, scTenifoldKnn, pySCENIC, mediation analysis), metabolic flux analysis (scFEA), and cell-cell communication analyses (CellChat, NicheNet, SecAct) were performed.
- **Results:** A novel schizophrenia risk gene was identified. In inhibitory neurons of a schizophrenia mouse model's prefrontal cortex (PFC), this risk gene drives synaptic vesicle dysfunction, metabolic reprogramming, and impaired neuron-microglia communication through a transcription factor regulatory axis.

Research Assistant

Shanghai Pudong New Area Mental Health Center, Tongji University School of Medicine, Shanghai, China

Jun. 2025 - Jan. 2026

Principal Investigator: Prof. Weijie Xie

- **Methods:** Integrative genetic analyses combining GWAS data with MR, TWAS, PWAS, and Bayesian colocalization were performed, followed by validation using single-cell datasets.
- **Results:** A cross-disorder genetic causal map of the PFC was constructed, elucidating the genetic basis of comorbidity mechanisms in psychiatric disorders.

Research Assistant

Shanghai Pudong New Area Mental Health Center, Tongji University School of Medicine, Shanghai, China

Aug 2024 - May 2025

Principal Investigator: Prof. Weijie Xie

- **Methods:** Single-nucleus RNA-seq data from the dorsolateral PFC of human Major Depressive Disorder (MDD) patients and single-cell data from a mouse depression model were integrated. Transcriptional regulatory network analysis (pySCENIC), metabolic pathway analysis (scMetabolism), differentiation trajectory inference (CytoTRACE/Monocle), 69 machine learning models, MR, and cell-cell communication analysis (CellChat) were applied to systematically characterize the functional features of microglial subtypes and their interactions with inhibitory neuron subtypes.
- **Results:** Three functionally distinct microglial subtypes were identified, with MG_2

defined as a pathogenic driver-like phenotype and MG_3 as a damage executor-like phenotype. It was predicted that in MDD, protective signaling axes (GAS6-TAM, VIP-VIPR1) were impaired, whereas pro-inflammatory pathways (TAC, VISFATIN) were aberrantly activated, establishing a selective bidirectional crosstalk network between microglial subtypes and inhibitory neuron subtypes.

Research Assistant

Shanghai Mental Health Center, Shanghai, China

Jul. 2023 – Jul. 2024

Principal Investigator: Assoc. Prof. Jiaoqiong Guan

- **Methods:** Performed integrative bioinformatics analyses (e.g., WGCNA, Machine Learning) and validation experiments using the Chronic Unpredictable Mild Stress model, with molecular techniques (Immunohistochemistry and Western Blot).
- **Results:** Identified an immune target that may underlie the shared immune pathogenesis which may drive the comorbidity progression of Systemic Lupus Erythematosus and MDD, with potential for clinical translation.

Master's Student Researcher (Focused on Microglia in Depressive-like Behaviors)

Xuzhou Medical University, Xuzhou, China

Sep 2020 - Jun 2023

Principal Investigator: Prof. Wuyang Wang (Supervisor)

- **Methods:** Applied *in vivo* (LPS / Chronic Restraint Stress mouse model, stereotaxic surgery) and *in vitro* (primary microglia isolation and culture) techniques.
- **Results:** Characterized a novel microglial lysosomal molecular mechanism implicated in the pathophysiology of depressive-like models and proposed a potential targeted intervention.

PUBLICATIONS

- **Qingqing Li, et al.** In Silico Integration of Genetics and Single-Cell Multi-Omics Nominates a Novel Risk Gene in Inhibitory Neuron Dysfunction in Schizophrenia. (Submitted)
- **Qingqing Li, et al.** Mendelian Randomization Analysis of Multiregional Psychiatric Disorders. (Submitted)
- **Qingqing Li, et al.** Single-Cell Dissection of Microglial Heterogeneity Reveals Subtype-Driven Immune-Metabolic Crosstalk in Major Depressive Disorder. *Current Neuropharmacology*. (Accepted, JCR Q1)
- **Qingqing Li, et al.** Microglia sing the prelude of neuroinflammation-associated depression. *Molecular Neurobiology*. 2025 Apr;62(4):5311-5332. (JCR Q1)
- Yanhong Xing, Meng-Meng Wang, Feifei Zhang, Tianli Xin, Xinyan Wang, Rong Chen, Zhongheng Sui, Yawei Dong, Dongxue Xu, Xingyu Qian, Qixia Lu, **Qingqing Li, et al.** Lysosomes finely control macrophage inflammatory function via regulating the release of lysosomal Fe²⁺ through TRPML1 channel. *Nature Communications*. 2025 Jan 24;16(1):985. (JCR Q1)
- Jiansong Qi, **Qingqing Li, et al.** MCOLN1/TRPML1 in the lysosome: a promising target for autophagy modulation in diverse diseases. *Autophagy*. 2024 Aug;20(8):1712-1722. (JCR Q1)

SKILLS

- **Wet-Lab Techniques:** qPCR, Western Blot, Immunofluorescence/Immunohistochemistry, ELISA, Golgi Staining, Gel Electrophoresis, Cell Culture, Animal Models.
- **Programming languages:** R, Python.
- **Bioinformatics:** Single-cell Analysis, Spatial Transcriptomics, DNA Methylation Analysis.
- **Statistical Genetics:** Mendelian Randomization, TWAS/PWAS, Colocalization.
- **Operating System:** Linux (command line, scripting).
- **Software:** EndNote, GraphPad Prism, ImageJ, Imaris, Adobe Illustrator & Photoshop.
- **Languages:** Mandarin Chinese (Native), English (**IELTS 6.5**).

ACADEMIC SERVICE

- Peer Reviewer for ***Molecular Neurobiology, Cellular and Molecular Neurobiology,*** and ***Scientific Reports.***

2024-Present

HONORS

- Postgraduate Scholarship, Xuzhou Medical University.

2020-2023